ZS-M4

Spectral Bridge & Transfer Operator

Q=11 Transfer Operator, Berry–Keating Bridge, and Prime-Resonance Diagnostics

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Verification: 25/25 PASS | Zero Free Parameters

§0. Abstract

We construct a finite-dimensional transfer operator on the Q=11 Z-Spin register equipped with a Z_2 seam involution J, and test its spectral diagnostics against non-trivial zeros of the Riemann zeta function. The operator's spectral determinant $|\det(I-L_s)|^2$ robustly separates Riemann zero heights from midpoint controls (Cohen's $d = 2.44$, $p < 0.002$), while composite-gate and random-phase negative controls remain non-significant.

The Berry–Keating bridge is established via five locking identities (L1–L5) at the i-tetration fixed point $z^* = i^{z^*}$, connecting Z-Spin's cross-coupling structure to the H=xp paradigm. The seam involution implements the functional-equation symmetry $s \leftrightarrow 1-s$ at the operator level ($\varepsilon_J = 0$ to machine precision).

Three honest falsifications are documented: (i) no simple A-based gap law exists; (ii) completed determinant $|D\xi|^2$ loses discriminating power; (iii) eigenvalue spacing is Poisson, not GUE. The midpoint bridge MAE ≈ 0.46 remains an open problem. This paper does NOT claim a proof of the Riemann Hypothesis.

Epistemic Status Legend

Status	Definition
PROVEN	Follows from standard mathematics alone. Machine-verifiable.
DERIVED	Follows from Z-Spin action + standard physics. Zero free parameters.
TESTABLE	Well-defined prediction/diagnostic awaiting further data or computation.
HYPOTHESIS	Structural parallel or conjecture. Derivation chain incomplete.
FALSIFIED	Tested and failed. Documented honestly as negative result.
CONSISTENT	Compatible with framework structure at finite Q. Not independently derived.
OPEN	Well-posed problem without current resolution.
NON-CLAIM	Explicitly not asserted. Documented to prevent overclaim.
CONTROL	Negative control test confirming specificity of positive results.

§1. Introduction

1.1 Scope and NON-CLAIM

ZS-M4 bridges the Z-Spin framework to the Hilbert–Pólya program for the Riemann Hypothesis. We present diagnostics and structural bridges, not a proof. Every claim carries an epistemic tag; every numerical pattern is tested against negative controls.

1.2 Locked Inputs

Input	Value	Source
A = 35/437	0.080092	ZS-F2 v1.0
(Z,X,Y) = (2,3,6), Q = 11	Slot register	ZS-F5 v1.0
z* = i^{z*}	0.43828 + 0.36059i	ZS-M1 v1.0
η_topo = $\|z^*\|^2$	0.32212	ZS-M1 v1.0
J: $\|j\rangle \to \|Q-1-j\rangle$	Z_2 seam involution	ZS-F5 v1.0

1.3 Epistemic Position

PROVEN: L1–L5 identities, $J^2=I$, JW_pJ=W_p*, seam consistency ε_J=0, functional equation Dξ(s)=Dξ(1−s). DERIVED: Spectral determinant D(s), completion factor B(s), ||L_s||≤1. TESTABLE: Spectral det separation d>1.9, negative control integrity. HYPOTHESIS: Berry–Keating connection, Hilbert–Pólya operator. FALSIFIED: A-based gap law, completed det discrimination. OPEN: GUE emergence (Q→∞), midpoint bridge (MAE≈0.46), self-adjoint extension.

§2. Berry–Keating Bridge: Locking Identities

At the i-tetration fixed point z* = i^{z*} (ZS-M1 v1.0), five identities hold to machine precision:

$$L1: \arg(z^*) = x^* \cdot (\pi/2) \quad (1)$$
$$L2: x^* = |z^*|\cos(\arg),\ y^* = |z^*|\sin(\arg) \quad (2)$$
$$L3: |z^*| = \exp(-y^* \cdot \pi/2) \quad (3)$$
$$L4: y^*/x^* = \tan(x^* \cdot \pi/2) \quad (4)$$
$$L5: |i^{z^*} - z^*| = 0 \quad (5)$$

Identity	Residual	Status
L1	1.11×10^{-16}	**PROVEN**
L2x	0.00	**PROVEN**
L2y	5.55×10^{-17}	**PROVEN**
L3	0.00	**PROVEN**
L4	1.11×10^{-16}	**PROVEN**

L5	1.11×10^{-16}	**PROVEN**

Bridge interpretation (HYPOTHESIS): The cross-coupling at z* produces a "roles crossed" mapping — the boost-like variable controls phase while the rotation-like variable controls magnitude — mirroring Berry–Keating's H=xp structure. This is a structural parallel, not a derivation.

§3. Q=11 Transfer Operator

3.1 Slot Space and Seam Involution

Computational space $\mathbb{C}^{11}$ with basis $|j\rangle$, j = 0,...,10.

$$J|j\rangle = |10-j\rangle,\ \ J^2 = I \quad (6)$$

Verified: $||J^2-I|| = 0$ to machine precision. [PROVEN]

3.2 Prime Gates

$$W_p = diag(exp(2\pi i(j-5)/p)),\ \ j = 0,...,10 \quad (7)$$

J-compatibility: J W_p J = W_p* for all primes. max||JW_pJ − W_p*|| = 0 across 80 primes. [PROVEN] (8)

3.3 Normalized Transfer Operator

$$L_s = (\Sigma_{p\leq P}\ p^{-s}\ W_p) / (\Sigma_{p\leq P}\ p^{-1/2}) \quad (9)$$

Evaluated on the critical line s = 1/2 + it. Computational setting: P = 80 primes (up to p = 409). This is a benchmark cutoff, not a free parameter. ||L_s||_op ≤ 1. [DERIVED]

3.4 Spectral Determinant

$$D(s) = det(I - L_s) \quad (10)$$

$$D\xi(s) = ½(B(s)D(s) + B(1-s)D(1-s)) \quad (11)$$

B(s) = ½s(s−1)π^{−s/2}Γ(s/2). By construction: Dξ(s) = Dξ(1−s). [PROVEN by construction]

§4. Seam Consistency

$$\varepsilon_J(t) = ||L_{1-s} - J\ L_s\dagger\ J||_F / ||L_{1-s}||_F \quad (12)$$

$\varepsilon_J = 0$ at 10 zero heights, 10 midpoints, and 10 random t values. The seam involution is exact (algebraic identity from J-compatibility). Seam reality: $|\mathrm{Im}\, D\xi|/|\mathrm{Re}\, D\xi|$ at zeros: mean = 1.05×10^{-16}. [PROVEN]

§5. Spectral Determinant Separation

$|\det(I-L_s)|^2$ at 20 zero heights vs 19 midpoints: Mean(zeros) = 5.396, Mean(mids) = 1.594, Cohen's d = 2.44, permutation p = 0.0002 (5000-shuffle estimate). At N=120 (full dataset): d = 1.96, $p < 10^{-6}$. [TESTABLE]

5.1 Prime-Phase Resonance

$$S(t) = |(1/N_p)\, \Sigma_{p\in P} \exp(it\cdot\log p)| \quad (13)$$

S(zeros) = 0.1270, S(mids) = 0.0502. Enhancement at zero heights. [HYPOTHESIS diagnostic]

§6. Negative Controls

Composite gates: d = 0.89 < 2.44 (prime). Weaker separation supports prime-specificity. [CONTROL]
Random phases: $\varepsilon_J \approx 1.4$ (vs 0 for primes). Seam consistency destroyed. [CONTROL]

§7. Honest Falsifications

F30-5: No simple A-based gap law exists. [FALSIFIED] F30-4: Eigenvalue spacing is Poisson, not GUE (consistent with finite Q=11). [CONSISTENT] Completed $|D\xi|^2$ does not discriminate zeros from midpoints (B(s) washes out signal). [FALSIFIED] Stage34 shows gap-matched composites can partially mimic separation. [DOCUMENTED]

7.1 What ZS-M4 Does NOT Establish

(a) The midpoint bridge is not a precision zero predictor (MAE ≈ 0.46). (b) The operator does not reproduce GUE statistics. (c) The completed determinant $D\xi$ does not discriminate. (d) Prime-specificity is limited.

§8. Open Problems

O1 (GUE emergence): $Q \to \infty$ limit → GUE?

O2 (Midpoint bridge): Coupled observable with MAE < 0.1?

O3 (Hilbert–Pólya): Self-adjoint extension reproducing ζ zeros?

§9. Claims

ID	Statement	Status
C1	L1–L5 at z* (5 identities)	**PROVEN**
C2	$J^2=I$ on $\mathbb{C}^{11}$	**PROVEN**
C3	JW_pJ = W_p* for all primes	**PROVEN**
C4	ε_J = 0 (seam consistency)	**PROVEN**
C5	$D\xi(s) = D\xi(1-s)$	**PROVEN**
C6	$\|\det\|^2$ separates zeros (d=2.44)	**TESTABLE**
C7	Composite NC weaker (d=0.89)	**CONTROL**
C8	No A-based gap law	**FALSIFIED (honest)**
C9	$\|D\xi\|^2$ non-discriminating	**FALSIFIED (honest)**
C10	NOT an RH proof	**NON-CLAIM**

§10. Verification Suite

Category	Tests	Pass	Scope
A: Locking identities L1–L5	5	5	Fixed-point residuals
B: Transfer operator	5	5	J^2, W_p, J-compat, ‖L_s‖, shape
C: Seam consistency	5	5	ε_J at zeros/mids/random, Dξ, reality
D: Spectral separation	5	5	$\|\det\|^2$, Cohen's d, perm, random, S(t)
E: Honest falsifications	5	5	Gap law, Poisson, $\|D\xi\|^2$, composite NC, NON-CLAIM
TOTAL	**25**	**25**	100% pass rate

Required dependencies: numpy, scipy, mpmath. The suite will not run without mpmath.

§11. Conclusion

The Q=11 transfer operator with Z_2 seam involution provides a well-defined, falsifiable diagnostic bridge between Z-Spin cosmology and the Hilbert–Pólya program. The spectral determinant $|\det(I-L_s)|^2$ robustly separates Riemann zero heights from controls ($d = 2.44$, $p = 0.002$). The seam involution enforces functional-equation symmetry exactly ($\varepsilon_J = 0$). Three honest falsifications (gap law, completed determinant, GUE statistics) are documented transparently. This paper makes no claim toward a proof of the Riemann Hypothesis.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite uses mpmath (50-digit) for z* and Γ(s/2); numpy/scipy double precision for matrix operations. Code is publicly available.

Appendix

A.1 Relation to ZS-M1 through ZS-M3

Paper	Interface	Status
ZS-M1 v1.0	z* = i^{z*}, L1–L5	Direct input ✓
ZS-M2 v1.0	X-Z-Y sector structure	J implements X↔Y exchange ✓
ZS-M3 v1.0	Z_2 seam involution	J same structure as κ=4 witness ✓
ZS-F5 v1.0	Q=11, (Z,X,Y)=(2,3,6)	Register dimension ✓

A.2 Key Numerical Values

Quantity	Value
z*	0.43828 + 0.36059i
|z*|	0.56756
η_topo = |z*|²	0.32212
||L_s||_op (at γ_1)	0.317
Cohen's d (N=20)	2.44
Permutation p (N=20)	0.0002
ε_J (seam error)	0.00 (exact)
d (composite NC)	0.89

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Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v2.0.0.)